

The G/L/C/M classification and shadow tables

Raez Lorgat

Remark 0.1 (Three independent depths). The inequalities $p_{\max} \neq k_{\max} \neq r_{\max}$ separate OPE pole order, collision depth, and shadow depth. The shadow classification is the stratification by the third invariant; the operator-order trichotomy is the stratification by the first two.

I The Four Shadow Classes

The single-line Riccati dichotomy partitions noncritical one-dimensional primary lines into the three rank-one outcomes G, L, and M. The fourth standard shadow class, C, is produced by charged-stratum separation in the bc and $\beta\gamma$ families. The rank-one outcomes are determined by two invariants of the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$: the cubic coefficient $\alpha = S_3$ and the critical discriminant $\Delta = 8\kappa S_4$.

Definition 1.1 (Shadow classes). Let $\mathcal{A}$ be a chirally Koszul algebra with shadow metric Q_L on a one-dimensional primary line L . The *shadow class* of $\mathcal{A}$ along L is:

Class	α	Δ	$r_{\max}$	<i>mechanism</i>
G (Gaussian)	0	0	2	<i>constant metric</i>
L (Lie/tree)	$\neq 0$	0	3	<i>square metric</i>
C (Contact)	0	line data	4	<i>charged contact stratum</i>
M (Mixed)	$\neq 0$	$\neq 0$	∞	<i>irreducible quadratic</i>

Here $r_{\max}$ is the *shadow depth*: the largest degree r for which the shadow coefficient $S_r \neq 0$. Class C is the finite contact stratum of the standard bc_λ and $\beta\gamma_\lambda$ families, where $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$ on the standard conformal-weight channel.

The classification is exhaustive: every entry in Table 1 falls into exactly one class, and every class is realized by at least one standard family. The mechanism is algebraic: the shadow generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is algebraic of degree 2 on each rank-one primary line, and its Taylor coefficients S_r are computed from the convolution recursion $H(t)^2 = t^4 Q_L(t)$, where $H(t) = \sum_{r \geq 2} r S_r t^r$ and $S_r = a_{r-2}/r$ with a_n the Taylor coefficients of $\sqrt{Q_L}$. When Q_L is a perfect square ($\Delta = 0$), the square root terminates; when Q_L is irreducible ($\Delta \neq 0$), the Taylor expansion is infinite.

1.1 Class G: Gaussian ($r_{\max} = 2$)

Definition 1.2. A chirally Koszul algebra $\mathcal{A}$ is *Gaussian* if $S_r(\mathcal{A}) = 0$ for all $r \geq 3$. Equivalently, $\alpha = 0$ and $\Delta = 0$, so the shadow metric $Q_L(t) = 4\kappa^2$ is constant and the shadow obstruction tower terminates at the scalar curvature.

Examples. Heisenberg $\mathcal{H}_k$ with $\kappa = k$; lattice VOAs V_Λ with $\kappa = \text{rank}(\Lambda)$; free fermion ψ with $\kappa = 1/4$.

Characterization. Class G algebras are freely generated as chiral algebras. Their bar complex is quadratic (the differential has no higher-order terms), and the OPE on each primary line is abelian. The full obstruction is captured by a single scalar:

$$\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda,$$

where η is the cyclic direction and $\Lambda \in H^*(\tilde{\mathcal{M}}_g)$ is the tautological class. All genus- g data is controlled by $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (uniform-weight).

1.2 Class L: Lie/tree ($r_{\max} = 3$)

Definition 1.3. A chirally Koszul algebra $\mathcal{A}$ is *Lie/tree* if $\alpha \neq 0$ and $\Delta = 0$, so that $S_3 \neq 0$ but $S_r = 0$ for all $r \geq 4$. The shadow metric is a perfect square: $Q_L(t) = (2\kappa + 3\alpha t)^2$.

Examples. Affine Kac–Moody $V_k(\mathfrak{g})$ for every simple Lie algebra $\mathfrak{g}$ and every non-critical level $k \neq -h^\vee$: type A ($\mathfrak{sl}_N$), types B, C, D (orthogonal and symplectic), exceptional types E_6, E_7, E_8, G_2, F_4 . For all of these:

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g}) (k + h^\vee)}{2h^\vee}.$$

Characterization. The nonzero cubic shadow $S_3 = \alpha$ arises from the Lie bracket (structure constants f_{abc}), while the Jacobi identity forces the quartic shadow $S_4 = 0$ on the primary line. In trace form,

$$r_{\text{tr}}^{\text{KM}}(z) = k \Omega_{\text{tr}}/z.$$

The KZ-normalized connection uses

$$r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k + h^\vee)z), \quad \Omega_{\text{KZ}} = 2h^\vee \Omega_{\text{tr}}.$$

These are conventionally related normalizations with different k -dependence. The tree-level graph sum terminates at trivalent vertices.

1.3 Class C: Contact ($r_{\max} = 4$)

Definition 1.4. A chirally Koszul algebra $\mathcal{A}$ is *contact* if $S_3 = 0$, $S_4 \neq 0$, and $S_r = 0$ for $r \geq 5$ on the standard contact channel. In the verified standard families this is the conformal-weight channel of bc_λ and $\beta\gamma_\lambda$, where $S_4 = -5/12$.

Mechanism. Class C is a finite contact phenomenon external to the rank-one Riccati dichotomy. The weight-changing line is abelian, while the standard conformal-weight channel carries the quartic contact coefficient. Rank-one rigidity kills the next charged target, so the tower stops at degree 4.

Examples. The $\beta\gamma$ system at conformal weight λ , with

$$\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1 = \frac{1}{2} c(\beta\gamma_\lambda),$$

and its Koszul dual, the bc system at the same weight. At the standard weight $\lambda = 0$ or $\lambda = 1$: $c = 2$ and $\kappa = 1$.

1.4 Class FF: Feigin–Frenkel locus ($\kappa = 0$)

The four classes G, L, C, M stratify the non-critical locus $\kappa(\mathcal{A}) \neq 0$. A fifth class appears at the boundary $\kappa = 0$: the *Feigin–Frenkel locus* where the Sugawara topologisation fails.

Definition 1.5 (Class FF). A chirally Koszul algebra $\mathcal{A}$ belongs to class FF if it lies at a critical Feigin–Frenkel centre or coset locus where the Sugawara topologisation degenerates. The vanishing $\kappa(\mathcal{A}) = 0$ is a necessary scalar signal in the affine examples; the class itself is controlled by the Feigin–Frenkel coset data $\text{coset}_{FF}(\mathcal{A})$.

Examples. The critical-level affine algebra $V_{-h^\vee}(\mathfrak{g})$ for every simple $\mathfrak{g}$ (so $V_{-2}(\mathfrak{sl}_2), V_{-3}(\mathfrak{sl}_3), \dots$); its commutant lies in the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}})$, polynomial in infinitely many Sugawara-like generators. The Koszul dual of class L under the involution $k \mapsto -k - 2h^\vee$ passes through class FF at the fixed point $k = -h^\vee$. Critical $\mathcal{W}_N$ (at the Feigin–Frenkel image of the affine critical level) is also class FF.

Characterization. At the Feigin–Frenkel critical locus the conformal vector degenerates: the Sugawara construction $T = (k + h^\vee)^{-1} : J^a J^a :$ has a pole, and topologisation $\text{SC}^{\text{ch}, \text{top}} \rightarrow E_3^{\text{top}}$ fails. The coset slot of the canonical fingerprint switches from $(\text{Com}(\mathcal{H}_\kappa, \mathcal{A}), \mathcal{A}/\text{Com})$ to the Feigin–Frenkel coset $\text{coset}_{FF}(\mathcal{A}) \subset \mathfrak{z}(\widehat{\mathfrak{g}})$. Class FF is the degeneration locus for the ambient shadow invariants. On this locus the controlling datum is the Feigin–Frenkel coset data instead of (α, Δ) .

Remark 1.6 (FF as a fifth companion class). The four classes G, L, C, M are determined by the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$, which degenerates at $\kappa = 0$: the leading constant term vanishes, the Riccati algebraicity argument loses its pivot, and the tower of $\Theta_{\mathcal{A}}^{\leq r}$ must be read from a different source. Together with G, L, C, M , this companion locus gives the five-class fingerprint stratification used in the standard landscape.

1.5 Class M: Mixed ($r_{\max} = \infty$)

Definition 1.7. A chirally Koszul algebra $\mathcal{A}$ is *mixed* if $\alpha \neq 0$ and $\Delta \neq 0$, so the shadow metric Q_L is an irreducible quadratic in t and the shadow obstruction tower does not terminate.

Examples. Virasoro Vir_c with $\kappa = c/2$; $\mathcal{W}$ -algebras $\mathcal{W}_N$ with $\kappa = (H_N - 1)c$ where $H_N = \sum_{j=1}^N 1/j$; Drinfeld–Sokolov reductions of affine Kac–Moody algebras.

Characterization. The shadow growth rate

$$\rho(\mathcal{A}) = \frac{\sqrt{9\alpha^2 + 16\kappa S_4}}{2|\kappa|}$$

controls the asymptotic behaviour of S_r : the coefficients satisfy $S_r \sim C(\mathcal{A}) \rho(\mathcal{A})^r r^{-5/2} \cos(r\theta + \phi)$ for a branch-point argument θ determined by the shadow metric. The critical central charge c^* where $\rho(\text{Vir}_c) = 1$ is the unique positive real root of the cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \quad c^* \approx 6.125. \quad (1)$$

For $c > c^*$ the tower converges ($\rho < 1$), and for $c < c^*$ it diverges ($\rho > 1$). At the self-dual point $c = 13$: $\rho(\text{Vir}_{13}) \approx 0.467$.

The Virasoro shadow obstruction tower has explicitly computed coefficients through degree 6:

$$\begin{aligned} S_2 &= \frac{c}{2}, & S_3 &= 2, & S_4 &= \frac{10}{c(5c + 22)}, \\ S_5 &= \frac{-48}{c^2(5c + 22)}, & S_6 &= \frac{4(240c + 1031)}{c^3(5c + 22)^2}. \end{aligned}$$

The identity $S_3 = 2$ is independent of c by direct residue extraction from the Virasoro OPE.

Remark 1.8 (Depth and Koszulness). Shadow depth $r_{\max}$ does *not* characterize chiral Koszulness. All four classes (Gaussian, $r_{\max} = 2$; Lie, $r_{\max} = 3$; contact, $r_{\max} = 4$; mixed, $r_{\max} = \infty$) consist of chirally Koszul algebras. Shadow depth classifies *complexity* within the Koszul world: the operadic complexity conjecture identifies $r_{\max}(\mathcal{A})$ with the $\mathcal{A}_\infty$ -depth and L_∞ -formality level.

Remark 1.9 (Total depth decomposition). The total shadow depth decomposes as $d(\mathcal{A}) = 1 + d_{\text{arith}} + d_{\text{alg}}$, where the algebraic depth $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ and depths ≥ 5 are purely arithmetic (cusp forms contribute beyond the algebraic engine).

1.6 The quadrichotomy with logarithmic and Wakimoto companions

The four classes G, L, C, M together with the Feigin–Frenkel locus FF form the fingerprint stratification of the non-logarithmic standard landscape. Two further labels occur on enlarged surfaces: logarithmic $\mathcal{W}(p)$ behaviour and Wakimoto free-field realisations. They are companion overlays that may overlap the five strata.

Definition 1.10 (Class M^* : logarithmic companion). A chirally Koszul algebra $\mathcal{A}$ carries the companion label M^* if it is logarithmic in the sense of Gurarie–Flohr [2, 5], i.e. the Zhu algebra $A(\mathcal{A})$ is finite-dimensional but the Hochschild Massey products $\langle \Omega, \Omega, \dots, \Omega \rangle$ carry amplitude on arbitrarily high tensor powers of the vacuum.

Examples. Triplet algebras $\mathcal{W}(p)$ for $p \geq 2$; the symplectic-fermion pair at central charge $c = -2$; logarithmic parafermion cosets.

Characterization. Class M^* algebras satisfy $\kappa \neq 0$ and share the mixed-shadow expansion $\alpha \neq 0$, $\Delta \neq 0$ of class M, but are distinguished by logarithmic amplitudes: Zhu-finite yet Massey-unbounded. The non-logarithmic shadow analysis applies on the formal-weight line; logarithmic corrections leave the algebraic invariants $\alpha = S_3$ and $\Delta = 8\kappa S_4$, which remain finite and computable, but they preclude tempering of the shadow tower.

Definition 1.11 (Class $\mathcal{W}$: Wakimoto stratum). A chirally Koszul algebra $\mathcal{A}$ belongs to class $\mathcal{W}$ if it arises as the screening-kernel of a rank- $(N - 1)$ Heisenberg through Feigin–Frenkel–Wakimoto realisation, $\mathcal{A} = \bigcap_i \ker Q_{\alpha_i} \subset \mathcal{H}^{\oplus(N-1)}$.

Examples. Principal $\mathcal{W}_N$ as a Wakimoto kernel inside $\mathcal{H}^{\oplus(N-1)}$ [3, 4]; affine $\widehat{\mathfrak{sl}}_2$ through the $\beta\gamma\phi$ Wakimoto realisation; admissible affine algebras through parabolic screenings [1, 6].

Characterization. Class $\mathcal{W}$ can overlap G, L, C, M : it is a presentational overlay. The ambient Heisenberg is class G , while the screening kernel may land in class L or class M after the BRST and screening-obstruction data are imposed. The label records the free-field presentation; Koszulity and Drinfeld coproduct descent require separate screening-kernel input.

Remark 1.12 (Fingerprint strata and overlays). The four shadow-depth classes G, L, C, M stratify the non-logarithmic non-critical locus by the algebraic invariants (α, Δ) of the shadow metric. Class FF marks the $\kappa = 0$ critical boundary. Class M^* records logarithmic amplitude behaviour orthogonal to (α, Δ) : a Zhu-finite class-M algebra is class-M, class- M^* , or both, depending on Massey boundedness [2, 5]. Class W records a presentation overlay: a class-L or class-M algebra may simultaneously be class- W when realised as a screening kernel. The stratification is five-class: G, L, C, M, FF ; M^* and W are companion labels.

1.7 Leading T-line exponential base

The leading large-central-charge coefficients on the Virasoro stress-tensor line have a sharp exponential base. This is a T-line statement, separate from exact finite- c radius questions for the full shadow series of an arbitrary class-M algebra.

Proposition 1.13 (Leading T-line exponential base $C_{\mathcal{A}}^T = 6$). *Let $\mathcal{A}$ be a non-logarithmic class-M standard family with a generic Virasoro stress-tensor line, and suppose the T-line dominance hypotheses hold. Let A_r^T denote the leading large- c coefficient of the T-line shadow coefficient S_r^T . Then*

$$C_{\mathcal{A}}^T := \limsup_{r \rightarrow \infty} |A_{r+1}^T / A_r^T| = 6.$$

For the Virasoro line the leading generating function has the logarithmic singularity

$$\Sigma_{\text{Vir}}^{T, \text{lead}}(z) = \frac{4z^3}{9} - \frac{z^2}{9} + \frac{z}{27} - \frac{\log(1+6z)}{162}.$$

Proof. The T-line large- c recursion is the Virasoro Riccati recursion with seed $S_3 = 2$. Solving the recursion gives the displayed logarithmic term, whose nearest singularity is at $z = -1/6$. The root test gives the exponential base 6 for the leading coefficients A_r^T . $\square$

For $\mathcal{W}_N$ the corresponding Vol II tempering exponent is a separate invariant. Under the harmonic tempering law it is

$$\beta_{\mathcal{W}_N} = 12(H_N - 1),$$

so $\beta_{\mathcal{W}_3} = 10$ and $\beta_{\mathcal{W}_4} = 13$. It is not the value given by the polynomial interpolant $(N+1)(N+2)/2$.

1.8 Picard–Fuchs spectral-curve presentation

The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ is a rank-one spectral datum; its square root $\sqrt{Q_L}$ is the Seiberg–Witten-type differential of a spectral cover. On non-logarithmic class M, the shadow generating function $\Sigma_{\mathcal{A}}(z)$ satisfies a Picard–Fuchs equation whose monodromy encodes the Borel geometry of the shadow tower.

Proposition 1.14 (Picard–Fuchs equation for the leading T-line). *Let $\mathcal{A}$ be class M with a generic Virasoro T-line and suppose the leading T-line dominance hypotheses of Proposition 1.13 hold. The leading T-line shadow series $\Sigma_{\mathcal{A}}^{T, \text{lead}}(z)$ satisfies a second-order Picard–Fuchs-type equation*

$$(1+6z)^2 (\Sigma_{\mathcal{A}}^{T, \text{lead}})'' + P_1(z) (\Sigma_{\mathcal{A}}^{T, \text{lead}})' + P_0(z) \Sigma_{\mathcal{A}}^{T, \text{lead}} = R_{\mathcal{A}}(z),$$

with $P_0, P_1, R_{\mathcal{A}}$ polynomials of degree at most 2 in z determined by the leading T-line Riccati data. The regular-singular point $z = -1/6$ is the leading T-line branch point.

Proof. Differentiate the logarithmic expression in Proposition 1.13. Clearing the denominator $(1 + 6z)^2$ gives the displayed inhomogeneous second-order equation with polynomial coefficients. The only singularity introduced by the logarithmic term in the leading T-line expression is $z = -1/6$. $\square$

2 Shadow Tables

The modular characteristic $\kappa(\mathcal{A})$ is the degree-2 projection of the universal MC element $\Theta_{\mathcal{A}}$, and the genus- g free energy is $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ for single-generator (uniform-weight) algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ are the Faber–Pandharipande intersection numbers.

2.1 Master shadow table

Table 1 records, for each standard family, the modular characteristic κ , the shadow class, the shadow depth $r_{\max}$, the cubic coefficient $\alpha = S_3$, the quartic coefficient S_4 , and the critical discriminant $\Delta = 8\kappa S_4$.

Table 1: Master shadow table: standard landscape

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	Class	$r_{\max}$	S_3	S_4	Δ
<i>Class G (Gaussian)</i>						
Heisenberg $\mathcal{H}_k$	k	G	2	0	0	0
Lattice V_Λ	$\text{rank}(\Lambda)$	G	2	0	0	0
Free fermion ψ	1/4	G	2	0	0	0
<i>Class L (Lie/tree)</i>						
$\widehat{\mathfrak{sl}}_2$ at level k	$3(k+2)/4$	L	3	$\neq 0$	0	0
$\widehat{\mathfrak{sl}}_N$ at level k	$(N^2-1)(k+N)/(2N)$	L	3	$\neq 0$	0	0
$\widehat{E}_6$ at level k	$13(k+12)/4$	L	3	$\neq 0$	0	0
$\widehat{E}_7$ at level k	$133(k+18)/36$	L	3	$\neq 0$	0	0
$\widehat{E}_8$ at level k	$62(k+30)/15$	L	3	$\neq 0$	0	0
<i>Class C (Contact)</i>						
bc ghosts (weight λ)	$-(6\lambda^2-6\lambda+1)$	C	4	0	$-\frac{5}{12}$	$\frac{10}{3}(6\lambda^2-6\lambda+1)$
$\beta\gamma$ (weight λ)	$6\lambda^2-6\lambda+1$	C	4	0	$-\frac{5}{12}$	$-\frac{10}{3}(6\lambda^2-6\lambda+1)$
<i>Class M (Mixed)</i>						
Vir_c	$c/2$	M	∞	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$
$\mathcal{W}_3$ at c	$5c/6$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_4$ at c	$13c/12$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_5$ at c	$77c/60$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_N$ at c (general)	$(H_N-1)c$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$

For bc_λ and $\beta\gamma_\lambda$, the class-C witness is the standard conformal-weight family: $S_2 = 6\lambda^2 - 6\lambda + 1$, $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$, with the sign of $\Delta = 8\kappa S_4$ set by the sign of κ .

In the table, $H_N = 1 + 1/2 + \dots + 1/N$ denotes the N -th harmonic number. The Virasoro quartic contact $S_4 = Q_{\text{Vir}}^{\text{contact}} = 10/[c(5c + 22)]$ is the unique quartic shadow on the T -line, proved by direct computation from the Virasoro OPE. The T -line shadow data ($S_3 = 2$, $S_4 = 10/[c(5c + 22)]$) is *universal*: it coincides for $\mathcal{W}_N$ at all $N \geq 2$, since every $\mathcal{W}_N$ contains the Virasoro subalgebra along the stress-tensor primary line. The additional primary lines (spin-3, spin-4, etc.) carry independent shadow data with separate quartic coefficients.

2.2 Exceptional and non-simply-laced types

Table 2 records the affine Kac–Moody data for exceptional and non-simply-laced types. All are class L.

Table 2: Shadow data for exceptional and non-simply-laced types

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\kappa(V_k(\mathfrak{g}))$	$c(V_k) + c(V_{k'})$	Class	$r_{\max}$
$B_2 \cong \mathfrak{so}_3$	10	3	$5(k+3)/3$	20	L	3
$C_2 \cong \mathfrak{sp}_4$	10	3	$5(k+3)/3$	20	L	3
G_2	14	4	$7(k+4)/4$	28	L	3
F_4	52	9	$26(k+9)/9$	104	L	3
E_6	78	12	$13(k+12)/4$	156	L	3
E_7	133	18	$133(k+18)/36$	266	L	3
E_8	248	30	$62(k+30)/15$	496	L	3

The kappa formula is universal: $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g}) (k + h^\vee)/(2h^\vee)$. The central charge sum $c + c' = 2 \dim(\mathfrak{g})$ under the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$, and the complementarity $\kappa + \kappa' = 0$ holds for all affine Kac–Moody algebras. For non-simply-laced types, the dual Coxeter number $h^\vee$ differs from the Coxeter number h ($h^\vee = 3$ for B_2 and C_2 , while $h = 4$). The shadow class is L regardless: the Jacobi identity kills the quartic on every primary line.

Remark 2.1 (B_2/C_2 isomorphism). The Lie algebras $B_2 = \mathfrak{so}_3$ and $C_2 = \mathfrak{sp}_4$ are isomorphic, sharing $\dim = 10$, $h^\vee = 3$, and $h = 4$. Their Cartan matrices are transposes of each other, but all shadow invariants coincide.

2.3 Complementarity table

[‡]The Koszul dual $\mathcal{H}_k^! = (\text{Sym}^{\text{ch}}(V^*[1]), m_o = -k\omega)$ has $\kappa(\mathcal{H}_k^!) = -k = \kappa(\mathcal{H}_{-k})$; its OPE structure differs from that of $\mathcal{H}_{-k}$. The central charge sum has no single-parameter-family meaning.

Here $K_N = 2(N-1)(2N^2+2N+1)$ is the Koszul conductor for $\mathcal{W}_N = \mathcal{W}(\mathfrak{sl}_N)$ (the sum $c + c'$ under the Feigin–Frenkel involution $k \mapsto -k - 2N$ on the underlying $\mathfrak{sl}_N$). The key structural distinction: for all affine Kac–Moody algebras, $\kappa + \kappa' = 0$ (anti-symmetry under the Feigin–Frenkel involution); for $\mathcal{W}$ -algebras, $\kappa + \kappa' = (H_N - 1) \cdot K_N \neq 0$. The Virasoro ($N = 2$) gives $\kappa + \kappa' = \frac{1}{2} \cdot 26 = 13$, a nonzero constant independent of c . The Koszul involution $c \mapsto 26 - c$ has a fixed point at $c = 13$, where $\kappa(\text{Vir}_{13}) = 13/2$ and Vir_{13} is self-dual under uncurved quadratic duality.

Table 3: Complementarity data: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$ and $c(\mathcal{A}) + c(\mathcal{A}^\dagger)$ under Koszul duality

Family $\mathcal{A}$	Koszul dual $\mathcal{A}^\dagger$	$c + c'$	$\kappa + \kappa'$	Self-dual point
$\mathcal{H}_k$	$\text{Sym}^{\text{ch}}(V^*)$	$\ddagger$	$\circ$	none
V_Λ (lattice)	$V_{\Lambda^\dagger}$	$\ddagger$	$\circ$	Λ self-dual
Free fermion	$\text{Sym}^{\text{ch}}(\gamma)$	$\circ$	$\circ$	none
$\beta\gamma_\lambda$	bc_λ	$\circ$	$\circ$	none
$V_k(\mathfrak{sl}_2)$	$V_{-k-4}(\mathfrak{sl}_2)$	6	$\circ$	$k = -2$ (critical)
$V_k(\mathfrak{sl}_N)$	$V_{-k-2N}(\mathfrak{sl}_N)$	$2(N^2-1)$	$\circ$	$k = -N$ (critical)
$V_k(\mathfrak{g})$ (general)	$V_{-k-2h^\vee}(\mathfrak{g})$	$2 \dim(\mathfrak{g})$	$\circ$	$k = -h^\vee$ (critical)
Vir_c	Vir_{26-c}	26	13	$c = 13$
$\mathcal{W}_3$ at c	$\mathcal{W}_3$ at $100-c$	100	$250/3$	$c = 50$
$\mathcal{W}_4$ at c	$\mathcal{W}_4$ at $246-c$	246	$533/2$	$c = 123$
$\mathcal{W}_5$ at c	$\mathcal{W}_5$ at $488-c$	488	$9394/15$	$c = 244$
$\mathcal{W}_N$ at c (general)	$\mathcal{W}_N$ at K_N-c	K_N	$(H_N-1) K_N$	$c = K_N/2$

2.4 Genus tables

The Faber–Pandharipande numbers and the genus expansion $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (uniform-weight).

 Table 4: Faber–Pandharipande intersection numbers λ_g^{FP}

g	λ_g^{FP}	Decimal
1	1/24	0.0416
2	7/5760	1.215×10^{-3}
3	31/967680	3.204×10^{-5}
4	127/154828800	8.201×10^{-7}
5	73/3503554560	2.084×10^{-8}

The formula is $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$ -th Bernoulli number. The generating function is $\sum_{g \geq 1} \lambda_g^{\text{FP}} \hbar^{2g} = \hat{A}(i\hbar) - 1$, where $\hat{A}(x) = (x/2)/\sinh(x/2)$ is the $\hat{A}$ -genus (so $\hat{A}(ix) = (x/2)/\sin(x/2)$). The numbers λ_g^{FP} are strictly positive, strictly decreasing, and decay like $(2\pi)^{-2g}$. The entries for $V_1(\mathfrak{sl}_3)$ use $\kappa = (8 \cdot 4)/(2 \cdot 3) = 16/3$. The Leech lattice (rank = 24) gives $F_1 = 1$, reflecting the Jacobi Δ -function origin of the Leech theta series.

2.5 Shadow growth rate table

The shadow growth rate $\rho = \sqrt{(9\alpha^2 + 16\kappa S_4)/(4\kappa^2)}$ determines the convergence of the shadow obstruction tower. For Virasoro, $\rho^2 = (180c + 872)/[c^2(5c + 22)]$. The critical central charge $c^* \approx 6.125$

Table 5: Genus expansion $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\text{FP}}$ (uniform-weight) for selected families

Algebra $\mathcal{A}$	κ	F_1	F_2	F_3
$\mathcal{H}_1$ (Heisenberg, $k=1$)	1	1/24	7/5760	31/967680
$V_1(\mathfrak{sl}_2)$	9/4	3/32	7/2560	31/430080
$V_1(\mathfrak{sl}_3)$	16/3	2/9	7/1080	31/181440
V_Λ (Leech, rank=24)	24	1	7/240	31/40320
$\beta\gamma$ (standard, $\lambda=1$)	1	1/24	7/5760	31/967680
Vir_1	1/2	1/48	7/11520	31/1935360
Vir_{13}	13/2	13/48	91/11520	403/1935360
Vir_{26}	13	13/24	91/5760	403/967680
$\mathcal{W}_3$ at $c=6$	5	5/24	7/1152	31/193536

Table 6: Shadow growth rate $\rho(\mathcal{A})$ for class M algebras (Virasoro at selected central charges)

c	κ	Q^{contact}	ρ	Regime
1/2 (Ising)	1/4	40/49	≈ 12.53	divergent
1	1/2	10/27	≈ 6.24	divergent
4	2	5/84	≈ 1.54	divergent
$c^* \approx 6.125$	≈ 3.06	≈ 0.031	1	critical
13 (self-dual)	13/2	10/1131	≈ 0.467	convergent
25	25/2	2/735	≈ 0.242	convergent
26 (bosonic string)	13	5/1976	≈ 0.232	convergent

(the positive real root of (1)) separates the convergent regime ($c > c^*$, tower summable) from the divergent regime ($c < c^*$, tower grows exponentially). Class G and class L algebras have $\rho = 0$ (the tower terminates, so there is no exponential growth).

2.6 Virasoro shadow obstruction tower table

Each S_r is a rational function of c with poles at $c = 0$ and $c = -22/5$; these are the two values where the quadratic Casimir $c(5c + 22)$ vanishes. The sign pattern $S_r > 0$ for even r and $S_r < 0$ for odd $r \geq 5$ (at $c > 0$) reflects the oscillatory decay $\cos(r\theta + \phi)$ from the complex branch points of the shadow generating function. All entries have been verified by two independent methods: closed-form derivation and the convolution recursion from $H^2 = t^4 Q_L$.

3 Structural Theorems

The classification is supported by four structural statements. Proposition 3.1 excludes intermediate algebraic depths; Proposition 3.2 identifies class **G** with SC-formality; and Proposition 3.4 produces class **C** by contact-channel separation. The Drinfeld–Sokolov conjecture isolates the class-raising functor

Table 7: Virasoro shadow coefficients S_r for $r = 2, \dots, 10$

r	$S_r(\text{Vir}_c)$
2	$c/2$
3	2
4	$10/[c(5c + 22)]$
5	$-48/[c^2(5c + 22)]$
6	$80(45c + 193)/[3c^3(5c + 22)^2]$
7	$-2880(15c + 61)/[7c^4(5c + 22)^2]$
8	$80(2025c^2 + 16470c + 33314)/[c^5(5c + 22)^3]$
9	$-1280(2025c^2 + 15570c + 29554)/[3c^6(5c + 22)^3]$
10	$256(91125c^3 + 1050975c^2 + 3989790c + 4969967)/[c^7(5c + 22)^4]$

under its generic-level hypotheses, and Proposition 3.8 gives the ordered classes their split direct-sum join.

3.1 The depth gap theorem

The Riccati equation leaves only three rank-one possibilities. The tower has support through S_2 in class **G**, through S_3 in class **L**, through a contact-channel S_4 in class **C**, and through infinitely many nonzero coefficients in class **M**. A quadratic Q_L is either a square or a non-square; its square root is respectively polynomial or an infinite binomial series. Thus the discriminant of Q_L determines the rank-one shadow depth, while class **C** is supplied by the separate standard contact channel.

Two distinct notions of depth appear in this analysis and must be kept separate. The algebraic depth $d_{\text{alg}}(\mathcal{A})$ counts independent shadow coefficients: $d_{\text{alg}} = 0$ if $S_r = 0$ for all $r \geq 3$, $d_{\text{alg}} = n$ if $S_r = 0$ for all $r \geq n + 3$, and $d_{\text{alg}} = \infty$ if infinitely many S_r are nonzero. The generating depth $d_{\text{gen}}(\mathcal{A})$ is the degree at which higher operations are recursively determined by lower ones. For the Virasoro algebra, $d_{\text{gen}} = 3$ (the cubic $S_3 = 2$ and quartic $S_4 = 10/[c(5c+22)]$ determine all S_r via the convolution recursion) but $d_{\text{alg}} = \infty$ (every S_r is nonzero). The recursion is finitely generated; the output has infinite support.

Proposition 3.1 (Depth gap). *Let $\mathcal{A}$ be a chirally Koszul algebra in the standard landscape with $\kappa(\mathcal{A}) \neq 0$. Then*

$$d_{\text{alg}}(\mathcal{A}) \in \{0, 1, 2, \infty\}.$$

*No finite value $d_{\text{alg}} \geq 3$ is realized. The four values correspond bijectively to the shadow classes **G**, **L**, **C**, **M**.*

Proof. The argument splits into a rank-one analysis on each primary line and a multi-sector extension.

Rank one. On a primary line L with $\kappa \neq 0$, the shadow generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is determined by (κ, α, S_4) , where the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ is the Gaussian decomposition and $\Delta = 8\kappa S_4$ is the critical discriminant.

If $\Delta = 0$ (equivalently $S_4 = 0$), then $Q_L(t) = (2\kappa + 3\alpha t)^2$ is a perfect square, and $H(t) = t^2(2\kappa + 3\alpha t)$ is a polynomial of degree 3. The shadow coefficients $S_r = 0$ for all $r \geq 4$, giving $d_{\text{alg}} \in \{0, 1\}$ according to whether $\alpha = 0$ (class **G**) or $\alpha \neq 0$ (class **L**).

If $\Delta \neq 0$ (equivalently $S_4 \neq 0$), then Q_L is irreducible over the rank-one coefficient field: the interaction correction $2\Delta t^2$ prevents $\sqrt{Q_L}$ from being polynomial. The Taylor expansion of $\sqrt{Q_L(t)}$

about $t = 0$ has infinitely many nonzero coefficients (the binomial series for $(1 + u)^{1/2}$ with $u = (6\alpha t + (9\alpha^2 + 2\Delta)t^2)/(4\kappa^2)$ has infinitely many nonzero terms when the coefficient of t^2 in u is nonzero). Therefore $S_r \neq 0$ for infinitely many r , and $d_{\text{alg}} = \infty$ (class **M**).

The gap at $d_{\text{alg}} = 3$ is now forced. Suppose $S_4 \neq 0$. The convolution recursion at degree 5 gives $S_5 = -(6/5)(1/\kappa) \alpha S_4$. If $\alpha \neq 0$: this is nonzero, so $S_5 \neq 0$. If $\alpha = 0$: then $S_5 = 0$, but the degree-6 recursion gives $S_6 \propto S_4^2/\kappa \neq 0$. In either case, $S_4 \neq 0$ propagates to infinitely many nonzero S_r : the closed form $H(t) = t^2 \sqrt{Q_L(t)}$ makes this manifest. Termination at any finite degree ≥ 5 is impossible.

Class **C** ($d_{\text{alg}} = 2$) occupies the contact channel where $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$ by a mechanism external to the rank-one Riccati analysis. See Proposition 3.4 below.

Multi-sector extension. For a multi-generator algebra with primary lines $L_1, \dots, L_n$, the cubic pump propagates: if $\alpha \neq 0$ on any mixed sector, the adjoint ad_{S_3} generates an infinite chain on every reachable sector, forcing $d_{\text{alg}} = \infty$. If all mixed cubics vanish, the sectors decouple and $d_{\text{alg}} \leq \max_i d_{\text{alg}}(L_i) \leq 2$. No intermediate value $d_{\text{alg}} = 3$ is accessible. $\square$

3.2 SC-formality characterises class **G**

Every algebra in the standard landscape is chirally Koszul: the bar spectral sequence collapses at E_2 , the bar cohomology concentrates in degree 1, and the transferred A_∞ operations on $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$ satisfy $m_k = 0$ for $k \geq 3$. Koszulness is common to all four shadow classes.

A stronger condition does. Swiss-cheese formality asks that the higher SC-operations on $\mathcal{A}$ *itself* vanish: $m_k^{\text{SC}} = 0$ for all $k \geq 3$. This is a condition on the algebra, distinct from a condition on its bar cohomology; the common notation m_k for both is a source of systematic confusion. The question is: which algebras satisfy this stronger condition?

The answer is a unique-survivor argument. Four shadow classes enter; one constraint eliminates three.

Proposition 3.2 (SC-formality iff class **G**). *Let $\mathcal{A}$ be a chiral algebra in the standard landscape. Then $\mathcal{A}$ is Swiss-cheese formal if and only if $\mathcal{A}$ belongs to class **G**.*

Proof. The forward direction is immediate. If $\mathcal{A}$ is class **G** (Heisenberg, lattice, free fermion), the current algebra bracket is central: $[a_m, a_n] = km \delta_{m+n,0}$ for Heisenberg, and the analogous statement for every other class **G** family. All nested brackets vanish, so $m_k^{\text{SC}} = 0$ for $k \geq 3$.

Conversely, suppose $\mathcal{A}$ is SC-formal. The cubic shadow S_3 is the normalized projection of the trilinear form

$$C(x, y, z) = \kappa(x, [y, z]),$$

where κ is the invariant bilinear form and $[-, -]$ the current algebra bracket. The chain of implications is:

$$\text{SC-formal} \implies m_3^{\text{SC}} = 0 \implies S_3 = 0 \implies C = 0.$$

The first arrow is the definition. The second holds because S_3 is the degree-3 shadow projection of m_3^{SC} . The third requires the key input: κ is nondegenerate on the standard landscape. With κ nondegenerate, $C = 0$ forces $[y, z] = 0$ for all y, z . The current algebra is abelian; an abelian current algebra at level k is Heisenberg; Heisenberg is class **G**.

Classes **L**, **C**, and **M** all have $S_3 \neq 0$ or carry structure (charged strata, nonlinear composites) incompatible with $m_3^{\text{SC}} = 0$. Each is eliminated. Only class **G** survives. $\square$

Remark 3.3 (The two meanings of formality). The distinction between $\mathcal{A}_\infty$ -formality of bar cohomology and SC-formality of the algebra itself is the central subtlety of the Koszulness programme. Every chirally Koszul algebra is $\mathcal{A}_\infty$ -formal on its bar cohomology; this is a consequence of the E_2 -collapse and holds in all four shadow classes. SC-formality is a strictly stronger condition, operative on the algebra rather than on its cohomological shadow. Proposition 3.2 identifies the precise locus where the two notions coincide: class **G**.

3.3 The betagamma contact mechanism

The single-line Riccati dichotomy produces three outcomes: $d_{\text{alg}} \in \{0, 1, \infty\}$. Class **C** ($d_{\text{alg}} = 2$) is absent from this list. Its existence requires a mechanism that the rank-one analysis misses: shadow data that lives on a charged stratum rather than on the primary line, and a termination mechanism that operates by dimensional obstruction rather than by algebraic identity.

The $\beta\gamma$ system is the algebra that exhibits this mechanism. It is the unique standard family where shadow depth exceeds collision depth: the OPE pole order is $p_{\text{max}} = 1$ (a single simple pole $\beta(z)\gamma(w) \sim 1/(z-w)$), the collision depth is $k_{\text{max}} = 0$ (no collision singularity), yet the shadow depth is $r_{\text{max}} = 4$. Three invariants that agree for every other standard family here separate: $p_{\text{max}} \neq k_{\text{max}} \neq r_{\text{max}}$. The mechanism is stratum separation: the quartic shadow lives on a charged stratum invisible to the single-line dichotomy, and the tower terminates there by rank-one rigidity.

Proposition 3.4 (Contact class via stratum separation). *Let $\beta\gamma_\lambda$ denote the $\beta\gamma$ system at conformal weight λ , with central charge $c(\beta\gamma_\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and $\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1$. Then:*

- (i) *On the weight-changing primary line, the OPE is abelian ($\alpha = 0$) and $\Delta = 0$: the single-line dichotomy gives $r_{\text{max}}|_L = 2$ (class **G** on this line).*
- (ii) *The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$ is nonzero and lives in the charged sector $\text{Def}_{\text{cyc}}^{(q)}$ with $q \neq 0$.*
- (iii) *The self-bracket $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$ has $\dim \text{Def}_{\text{cyc}}^{(2q)} = 0$ for rank-one systems: the quartic pump has zero target, and the tower terminates at $r_{\text{max}} = 4$.*
- (iv) *The shadow depth is $r_{\text{max}}(\beta\gamma_\lambda) = 4$ and $d_{\text{alg}}(\beta\gamma_\lambda) = 2$ for all λ .*

Proof. The primary-line computation gives (i), the charged-sector decomposition gives (ii), rank-one rigidity gives (iii), and (i)–(iii) imply (iv).

Item (i): invisibility on the primary line. The $\beta\gamma$ OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ is a simple pole. The self-OPEs $\beta(z)\beta(w)$ and $\gamma(z)\gamma(w)$ are regular. There is no current algebra bracket: the OPE on the weight-changing line is abelian, $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. The Riccati dichotomy applied to this line gives $d_{\text{alg}} = 0$. If the primary line were the whole story, $\beta\gamma$ would be class **G**.

Item (ii): birth of the quartic on the charged stratum. The weight-changing primary line is incomplete. The $\beta\gamma$ system has two generators of different conformal weight: the cyclic deformation complex decomposes by charge, and the charged sector $\text{Def}_{\text{cyc}}^{(q)}$ with $q \neq 0$ is invisible to the restriction $q = 0$ that defines the primary line. The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$ is the trilinear Massey product on this charged stratum. In the standard conformal-weight family the quartic coefficient is $S_4 = -5/12$ for all λ ; at $\lambda = 1/2$ the symplectic boson changes the parity grading, while the graded contact witness remains nonzero.

Item (iii): death of the quintic by rank-one rigidity. The quartic shadow, once born, must either propagate (producing $d_{\text{alg}} = \infty$, class **M**) or die (giving $d_{\text{alg}} = 2$, class **C**). The propagation mechanism

is the quartic pump: the self-bracket $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$. For rank-one systems (one generator in each charge sector), the target space has dimension zero: there is no field of charge $2q$ in a rank-one system. The quintic obstruction S_5 vanishes identically on the charged stratum; the pump has no room to operate. The tower terminates.

Item (iv). Items (i)–(iii) give $S_r = 0$ for $r \geq 5$ on every stratum and $S_4 \neq 0$ on the charged stratum: $r_{\max} = 4$, $d_{\text{alg}} = 2$. The result holds for all λ (the rank-one rigidity is independent of conformal weight), confirming class **C**. $\square$

Remark 3.5 (Why contact is rare). The mechanism that produces class **C** requires three simultaneous conditions: an abelian primary line ($\alpha = 0$), a nonzero quartic on a charged stratum ($\mathfrak{Q} \neq 0$), and rank-one rigidity to terminate the tower ($\dim \text{Def}^{(2q)} = 0$). The third condition fails for rank ≥ 2 : the target of the quartic pump acquires positive dimension, and the tower generically diverges. Among standard families, only the $\beta\gamma$ system and its Koszul dual bc satisfy all three conditions.

3.4 Shadow depth under Drinfeld–Sokolov reduction

The four shadow classes are static invariants of individual algebras. A natural question is how these invariants transform under the functors that connect families. Quantum Drinfeld–Sokolov reduction is the principal such functor: it sends an affine Kac–Moody algebra $V_k(\mathfrak{g})$ to a $\mathcal{W}$ -algebra $\mathcal{W}^k(\mathfrak{g}, f)$, parameterised by a choice of nilpotent $f \in \mathfrak{g}$. At $f = 0$ the reduction is trivial and the output is the input. At $f \neq 0$ the output is a genuinely different algebra: new composite fields appear (the Sugawara stress tensor T , higher-spin currents W_s), and these composites carry OPE data absent from the affine algebra.

The conjectural shadow-theoretic consequence is class escalation. An affine Kac–Moody algebra is class **L** ($r_{\max} = 3$): the Jacobi identity kills the quartic shadow on every primary line. For a generic nontrivial DS reduction, the stress-tensor line is required to carry a nonzero Virasoro quartic $S_4 = 10/[c(5c+22)]$, away from the singular central charges $c = 0, -22/5$. The depth gap would then force $r_{\max} = \infty$. Singular levels and degenerate central charges require separate computation.

Conjecture 3.6 (Shadow depth escalation under DS reduction). *Let $\mathfrak{g}$ be a simple Lie algebra, $f \in \mathfrak{g}$ a non-zero nilpotent, and let k be a non-critical level for which the stress-tensor line of $\mathcal{W}^k(\mathfrak{g}, f)$ is nondegenerate and has $c \notin \{0, -22/5\}$. Then*

$$r_{\max}(\mathcal{W}^k(\mathfrak{g}, f)) \geq r_{\max}(V_k(\mathfrak{g})) = 3.$$

*The conjecture asserts strictness for every non-zero nilpotent: $r_{\max}(\mathcal{W}^k(\mathfrak{g}, f)) = \infty$ (class **M**).*

The conjecture asserts that DS reduction is a one-way door in shadow complexity: once the nilpotent f is turned on, the tower remains infinite in the conjectural generic regime. The depth gap (Proposition 3.1) reduces the conjecture to a single computation: it suffices to show $S_4 \neq 0$ on the stress-tensor primary line of $\mathcal{W}^k(\mathfrak{g}, f)$. The infinite tower then follows automatically.

Remark 3.7 (Evidence). Three independent lines of evidence support the conjecture.

(a) *Principal reductions.* For $f = f_{\text{prin}}$ and the standard generic-level principal rows, the statement is proved: the principal $\mathcal{W}$ -algebra $\mathcal{W}_N = \mathcal{W}^k(\mathfrak{sl}_N, f_{\text{prin}})$ contains the Virasoro algebra as a subalgebra, and the Virasoro quartic $S_4 \neq 0$ on the T -line is a universal computation independent of $\mathfrak{g}$.

(b) *Non-principal reductions in type A.* Existing computations support the conjecture for several nilpotent orbits of $\mathfrak{sl}_N$ at non-critical levels k . The generic statement remains conjectural outside the verified OPE tables. The minimal nilpotent orbit of $\mathfrak{sl}_3$ (the Bershadsky–Polyakov algebra $\mathcal{W}^k(\mathfrak{sl}_3, f_{\min})$) is the simplest non-principal example: it has a single strong generator beyond the stress tensor, and $S_4 \neq 0$ is verified by direct OPE computation.

(c) *Structural mechanism.* Every non-trivial nilpotent f produces, via the BRST cohomology of the associated quantum DS complex, at least one composite strong generator whose self-OPE has pole order ≥ 4 . The pole order of the composite is bounded below by the Sugawara construction: the stress tensor T always appears as a strong generator, and $T(z)T(w)$ has a quartic pole. This quartic pole creates $S_4 \neq 0$ on the stress-tensor primary line, and the depth gap converts this single nonzero quartic into an infinite tower.

3.5 The join-semilattice of shadow classes

The four shadow classes are invariants of individual algebras. Their behaviour under algebraic operations requires a separate argument. The split operation considered here is the direct sum of cyclically admissible Lie conformal algebras with vanishing mixed OPE, whose modular envelopes split as completed tensor products.

The answer is the maximum, and the proof is that the shadow tower decomposes sector by sector when the cross-sector OPE is regular. This gives the four shadow classes the structure of a join-semilattice: a partially ordered set in which every pair of elements has a least upper bound, realised concretely by direct sum.

Proposition 3.8 (Shadow classes form a join-semilattice). *The four shadow classes $\{\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}\}$ form a join-semilattice under the total order $\mathbf{G} < \mathbf{L} < \mathbf{C} < \mathbf{M}$, where the join is realised by direct sum of chiral algebras:*

$$\text{class}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(\text{class}(\mathcal{A}_1), \text{class}(\mathcal{A}_2)).$$

Equivalently, the shadow depth satisfies $r_{\max}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(r_{\max}(\mathcal{A}_1), r_{\max}(\mathcal{A}_2))$.

Proof. Assume $\mathcal{A}_1$ and $\mathcal{A}_2$ are cyclically admissible Lie conformal algebras with vanishing mixed OPE and split modular envelopes. The OPE of $\mathcal{A}_1 \oplus \mathcal{A}_2$ decomposes into three sectors: pure $\mathcal{A}_1$, pure $\mathcal{A}_2$, and mixed. The mixed sector has regular OPE by hypothesis: fields from distinct factors are mutually local with no singular terms. Regular OPE contributes no shadow coefficients at any degree. The shadow tower of $\mathcal{A}_1 \oplus \mathcal{A}_2$ is therefore the direct sum of the towers of $\mathcal{A}_1$ and $\mathcal{A}_2$, and the shadow depth is the maximum of the two.

The decisive case is $\mathbf{L} \oplus \mathbf{L}$: does combining two affine Kac–Moody factors produce a quartic shadow? The answer is no. The quartic obstruction S_4 is intrinsic to each factor’s OPE. On the primary line of $V_k(\mathfrak{g}_1)$, the Jacobi identity kills S_4 ; on the primary line of $V_{k'}(\mathfrak{g}_2)$, the same. The mixed channel $\mathfrak{g}_1 \otimes \mathfrak{g}_2$ contributes no singular OPE terms, so no quartic shadow is born in the cross-sector. Therefore $\mathbf{L} \oplus \mathbf{L} = \mathbf{L}$: the Jacobi identity of each factor survives the direct sum. $\square$

Remark 3.9 (Monotonicity and its failure). The join-semilattice property is specific to direct sums: the functor $\oplus$ preserves the regularity of cross-sector OPE. Constructions that create new singular terms between sectors violate monotonicity and can raise the shadow class strictly. Two examples illustrate the failure.

Drinfeld–Sokolov reduction sends class **L** affine input to class **M** output in the verified principal rows and in the generic regime predicted by Conjecture 3.6. The BRST construction creates composite fields with new OPE singularities; these singularities produce the quartic shadow that the Jacobi identity had killed.

Coset construction: $\mathcal{A}/\mathcal{A}'$ can land in a higher class than either factor. The Goddard–Kent–Olive coset of $V_k(\mathfrak{sl}_2)$ by its Cartan subalgebra produces the parafermion algebra, whose shadow depth depends on the coset data rather than on the shadow classes of the two inputs.

Shadow depth is monotone under direct sum, anti-monotone under no operation, and unconstrained under quotient or reduction. The join-semilattice records the stable part of this structure; the unstable part is the subject of Conjecture 3.6.

References

- [1] T. Arakawa, *Representation theory of W -algebras*, Invent. Math. **169** (2007), no. 2, 219–320.
- [2] M. A. I. Flohr, *On modular invariant partition functions of conformal field theories with logarithmic operators*, Int. J. Mod. Phys. A **11** (1996), 4147–4172.
- [3] C. Faber and R. Pandharipande, *Hodge integrals, partition matrices, and the λ_g conjecture*, Ann. of Math. **157** (2003), no. 1, 97–124.
- [4] I. B. Frenkel, V. G. Kac, and M. Wakimoto, *Characters and fusion rules for W -algebras via quantized Drinfeld–Sokolov reduction*, Comm. Math. Phys. **147** (1992), no. 2, 295–328.
- [5] V. Gurarie, *Logarithmic operators in conformal field theory*, Nucl. Phys. B **410** (1993), 535–549.
- [6] V. G. Kac, S.-S. Roan, and M. Wakimoto, *Quantum reduction for affine superalgebras*, Comm. Math. Phys. **241** (2003), 307–342.